

Network Types — Guided Notes ANSWER KEY

Unit 1.2.2 | CompTIA Network+ N10-009 Obj. 1.2 | N10-008 1.2

For instructor use / distribute after students complete the notes.

Question 1

A _____ (LAN) covers a single building or small group of buildings and is typically owned by _____. A _____ (WAN) connects networks across large geographic distances and usually crosses infrastructure owned by _____.

Answer: Local Area Network; the organization; Wide Area Network; carriers or ISPs.

Question 2

Match each network type to its scope:

Personal Area Network (PAN) → _____

Campus Area Network (CAN) → _____

Metropolitan Area Network (MAN) → _____

Answer: PAN → ~1 meter, personal devices (Bluetooth, NFC); CAN → multiple buildings on a defined campus; MAN → city or large district.

Question 3

A WLAN provides the same connectivity as a wired LAN but uses _____ instead of copper or fiber. Two key tradeoffs versus wired Ethernet are _____ and _____.

Answer: Radio frequencies (802.11/Wi-Fi); shared medium / half-duplex performance; inherently less secure (signal is not contained inside a cable). Also acceptable: signal degradation with distance.

Question 4

MPLS routes WAN traffic using short _____ instead of full IP addresses. Businesses buy MPLS circuits for _____ and _____. The main disadvantage of MPLS compared to newer options is _____.

Answer: Path labels; guaranteed bandwidth (SLAs); predictable performance. Main disadvantage: high cost.

Question 5

SD-WAN places a _____ layer on top of multiple transport links (MPLS, broadband, LTE) and routes traffic based on _____. The business benefit over traditional MPLS is _____.

Answer: Software; real-time performance metrics; lower cost — commodity internet links can replace or supplement expensive MPLS circuits while maintaining application-aware routing.

Question 6

In a _____ network architecture, dedicated servers provide resources and clients consume them. In a _____ architecture, every device can act as both client and server. At approximately _____ devices, peer-to-peer becomes unmanageable.

Answer: Client-server; peer-to-peer (P2P); 10 devices.

Question 7

A SOHO network is typically a _____ design with one _____ and one _____ domain. The all-in-one device combines a router, switch, wireless access point, DHCP server, firewall, and _____ in a single box.

Answer: Flat; subnet; broadcast; NAT gateway.

Question 8 — Real World Application

REAL WORLD: A school's IoT security cameras have been compromised and are sending spam traffic. The cameras share the same network as student workstations and staff file servers. What network design change would contain the damage and prevent the cameras from reaching production systems? Explain the concept behind your solution.

Answer: Segment the IoT devices onto a separate VLAN or subnet, isolated from the production LAN. A compromised device on a different VLAN cannot reach the file server without traffic crossing a router, where a firewall policy can block it. Best practice: IoT VLAN with a deny-all-to-production rule, allowing only necessary outbound internet traffic for device management.